

Modification of iron uptake and lipid peroxidation by hypoxia, ascorbic acid, and α -tocopherol in iron-loaded rat myocardial cell cultures

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The ability of ascorbic acid, α -tocopherol, and hypoxia to modify iron uptake, chelation, and toxicity as manifested by the generation of malonyldialdehyde (MDA) was studied in myocardial cell cultures obtained from newborn rats. Exposure to 20 μ g/ml iron provided as ^{59}Fe -ferric ammonium citrate in serum-free Ham F-10 culture medium resulted in the accumulation of 39% of the iron within 24 hours and a 10- to 12-fold increase in cellular MDA. Hypoxia (1% oxygen) resulted in a more than twofold increase in iron uptake but only minor changes in cellular MDA concentrations. Ascorbic acid and α -tocopherol (1 mg/ml) had opposing effects on iron uptake and MDA production. Ascorbic acid reduced 24-hour iron uptake by 73% ($P < 0.001$) whereas α -tocopherol increased iron uptake by 19% ($P < 0.025$). In contrast, cellular MDA after iron loading increased by 86% with the addition of ascorbate, and was reduced by 75% with α -tocopherol ($P < 0.001$). The ratio of increase in cellular MDA relative to percent iron uptake (lipid peroxidation ratio) was 7.29 with iron loading plus ascorbate vs. 0.13 with iron loading plus α -tocopherol, a 56-fold difference between the two extremes. In vitro deferoxamine treatment for 3 hours resulted in a 53% reduction in the radioactive iron content of iron-loaded heart cells and a 40% reduction in MDA. Simultaneous deferoxamine and ascorbate or α -tocopherol treatment did not affect iron mobilization, but had a profound effect on MDA concentrations. Ascorbic acid prevented entirely the beneficial effect of deferoxamine on MDA concentrations in iron-loaded cells, whereas α -tocopherol potentiated the effect of deferoxamine. These observations may be pertinent to the aggravating effect of coexistent hypoxia on the severity of myocardial iron accumulation and the possible aggravation of myocardial damage by ascorbate therapy, and underline the need to explore the therapeutic potential of α -tocopherol in protecting tissues from the harmful effects of iron excess. (J LAB CLIN MED 1987;110:355-61)

Abbreviation: MDA = malonyldialdehyde

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Increased lipid peroxidation is an important manifestation of both chronic and acute iron intoxication.¹⁻⁸ Ferric iron is converted by superoxide into ferrous iron, which in turn catalyzes the conversion of hydrogen peroxide into the hydroxyl radical in the Haber-Weiss reaction.⁹⁻¹¹ This reaction is accelerated by ascorbic acid, a reducing agent.^{5,8,9,12} Hydroxyl radicals initiate a chain reaction of lipid peroxidation and membrane injury terminating in the production of

Table I. Effect of hypoxia on iron uptake and cellular MDA concentrations

Time (hr)	Iron loading			Iron loading plus hypoxia		
	n	⁵⁹ Fe (%)	MDA (nmol/mg)	n	⁵⁹ Fe (%)	MDA (nmol/mg)
0	4	0	0.283 ± 0.032	4	0	0.283 ± 0.032
0.25	4	4.3 ± 0.1	1.718 ± 0.167	4	8.5 ± 0.6	1.654 ± 0.077
0.5	4	4.2 ± 0.1	2.596 ± 0.269	4	11.2 ± 1.3	2.532 ± 0.199
1	4	6.3 ± 0.6	3.294 ± 0.365	4	16.5 ± 1.5	3.282 ± 0.269
3	4	15.0 ± 1.8	3.147 ± 0.244	4	23.5 ± 1.0	3.962 ± 0.160

Values are expressed as mean ± SD.

MDA. This chain reaction may be inhibited by the antioxidant action of α -tocopherol demonstrated both in vitro and in vivo.^{7,8} In a previous study¹³ we have shown that acute iron loading in cultured rat myocardial cells is associated with a rapid increase in lipid peroxidation as indicated by increased cellular MDA concentrations and that iron mobilization by deferoxamine results in normalization of cellular MDA in direct proportion to the amounts of iron removed. The objective of the present studies was to explore the capacity of ascorbic acid, α -tocopherol, and hypoxia to modify iron toxicity as manifested in the generation of MDA by iron-loaded cultured rat myocardial cells.

METHODS

Ham F-10 culture medium (Beth Haemek, Israel) supplemented with $\text{CaCl}_2 \cdot \text{H}_2\text{O}$ 135 mg/L, penicillin 200,000 U/L, streptomycin 0.2 mg/L, 10% horse serum, and 10% fetal bovine serum (GIBCO, Grand Island, N.Y.) was used as growth medium. For iron uptake studies, heterologous serum was replaced by serum-free medium. For mincing and washing the organs and for trypsinization, H solution was used. Trypsin (type III; Sigma Chemical Co., St. Louis) was dissolved in 0.1 wt/vol H solution.

Deferoxamine was obtained from Ciba-Geigy Corp., Summit, N.J. Ascorbic acid (Cereon; Assia, Ramat Gan, Israel) and α -tocopherol (Ephymal; Hoffmann-La Roche, Nutley, N.J.) were both used at a concentration of 1 mg/ml. Measurement of cellular MDA concentrations was performed by the method of Stocks and Dormandy.¹⁴

Preparation of isolated cells. Cultures from 1-day-old rats (Hebrew University strain) were obtained by a slight modification of previously published methods.¹⁵ The newborn rats were killed by decapitation. Hearts were removed aseptically, retaining the ventricles only. The excised hearts were placed in a Petri dish containing H solution. The organs were minced into very small fragments and washed twice during mincing. H solution was then replaced by trypsin solution and transferred to a 25 ml flask containing a Teflon-covered magnetic bar attached to the stopper. The volume of trypsin solution was 10 to 15 ml for 30 to 50 hearts. Trypsinization was performed at 32° to 35° C with a stirring rate of 150 to 200 rpm for 15 to 20 minutes. At the end of each trypsin-

ization, fragments were allowed to settle, the supernatant was removed, and another portion of trypsin was added. Supernatants from the first two or three trypsinizations, which contained mostly cell debris, red blood cells, and pericardial and endothelial cells, were discarded. Cells from subsequent trypsinizations were collected in 30 ml sterile tubes. A few milliliters of growth medium were added to stop the trypsin digestion, and cells were centrifuged at 1000 rpm. The pellet was resuspended in a small volume of growth medium. Trypsinizations were repeated until all fragments were dissociated.

Cell cultures. Cells from each tube were resuspended by repeated aspiration into a sterile pipette, and all fractions were combined into a sterile 250 ml flask (Nunc; Nunclon Delta, Herlev, Denmark) through a sterile mesh, to exclude explants. The pooled cells were diluted in growth medium to a density of 9×10^5 to 1×10^6 cells per milliliter and seeded into a 35 mm diameter Petri dish (Falcon 3001; Falcon Labware, Oxnard, Calif.). After seeding, Petri dishes were gently shaken horizontally to obtain a uniform distribution of cells. This concentration yielded after 24 to 36 hours an almost confluent layer at a final density of about 2×10^6 cells. Cultures were kept at 37° C in an atmosphere of 5% CO_2 and 95% air. Experiments were performed at 5 days of culture when >80% of the cells were beating myocardial cells. Continued viability of cultured iron-loaded cells was documented by supravital dye exclusion and by the absence of enzyme (lactate dehydrogenase) leakage into the culture medium after 24-hour iron loading. The pH of cultures was maintained at 7.4 ± 0.05 .

Radioactive iron labeling. $^{59}\text{FeCl}_3$ (specific activity 10 to 15 $\mu\text{Ci}/\mu\text{g}$; Amersham Radiochemical Centre, Amersham, England) was diluted in 0.005N HCl and mixed with sufficient sterile ferric ammonium citrate (BDH Fine Chemicals Ltd., Poole, England) to provide a concentration of 100 $\mu\text{g}/\text{ml}$. Preliminary studies have shown that 24 hours incubation of the cultures with 2.5, 5, 10, and 20 μg iron per milliliter resulted in mean ($\pm$ SD) cellular MDA concentrations of 0.788 ± 47 , 1.011 ± 34 , 1.476 ± 347 , and 3.107 ± 367 nmol/mg protein respectively. In all subsequent studies, final iron concentrations of 20 $\mu\text{g}/\text{ml}$ have been used. To 0.8 ml cells in culture medium, 0.2 ml of the 100 $\mu\text{g}/\text{ml}$ iron solution was added to provide a final iron concentration of 20 $\mu\text{g}/\text{ml}$. At the termination of incubation, culture plates were washed twice with 1 ml cold culture medium. The cells were scraped and transferred into counting tubes by means of a rubber

Table II. Effect of ascorbic acid and α -tocopherol on cellular iron uptake and MDA concentrations

Time (hr)	Iron loading	Iron loading plus ascorbate	Iron loading plus α -tocopherol
⁵⁹ Fe uptake (%)			
0	0	0	0
0.25	2.07 $\pm$ 0.16	1.94 $\pm$ 0.51	3.35 $\pm$ 0.09
1	4.32 $\pm$ 0.51	3.81 $\pm$ 0.41	6.31 $\pm$ 0.20
3	12.86 $\pm$ 0.37	6.31 $\pm$ 0.22	15.60 $\pm$ 2.22
24	32.90 $\pm$ 5.23	8.92 $\pm$ 0.38	39.12 $\pm$ 0.44
MDA (nmol/mg)			
0	0.276 $\pm$ 0.032	0.276 $\pm$ 0.032	0.276 $\pm$ 0.032
0.25	1.833 $\pm$ 0.333	4.647 $\pm$ 0.609	1.782 $\pm$ 0.115
1	2.500 $\pm$ 0.160	5.795 $\pm$ 0.263	1.474 $\pm$ 0.218
3	2.762 $\pm$ 0.128	7.282 $\pm$ 0.231	1.628 $\pm$ 0.218
24	2.339 $\pm$ 0.186	4.352 $\pm$ 0.494	0.583 $\pm$ 0.167

Each value represents mean $\pm$ SD of four studies.

policeman and resuspended in 0.5 ml culture medium. ⁵⁹Fe activity was determined in an automatic well scintillation counter (Auto-Gamma model 5360; Packard Instrument Co., Inc., Downers Grove, Ill.).

The paired t test was used for comparison of differences within the same experimental group and the unpaired t test for comparing differences between group means.

RESULTS

The effect of hypoxia on iron uptake and cellular MDA concentrations is described in Table I. The conditions of cell culture and concentrations of iron were identical in the two arms of the study, except for the replacement of 20% oxygen in iron-loaded controls by 1% oxygen in the hypoxic group starting at time 0 and throughout the 3 hours of study. Hypoxia was started simultaneously with the addition of radioactive iron-labeled ferric ammonium citrate at a concentration of 20 μ g/ml elemental iron.

At both normal and hypoxic conditions there was a gradual accumulation of iron in myocardial cells. Under hypoxic conditions iron uptake was more than twofold higher than at normal atmospheric conditions at all points examined ($P < 0.001$). With increasing cellular iron deposition there was a rapid initial increase in cellular MDA concentrations. However, unlike iron uptake, cellular MDA concentrations reached near-maximal levels within the first hour of iron loading with only a minor increase thereafter. In contrast to the accelerated rate of iron uptake in hypoxic cells, there was no difference in the MDA content of cells under normal or hypoxic conditions within the first hour of study, and only at 3 hours could a 26% increase in cellular MDA ($P < 0.001$) be demonstrated in hypoxic iron-loaded cells.

In the following studies the effect of ascorbic acid and α -tocopherol on the rate of cellular iron uptake and

MDA concentrations was studied (Table II). Both compounds were used at a concentration of 1 mg/ml added at time 0, simultaneously with the addition of radioactive iron-labeled ferric ammonium citrate at the concentrations indicated above.

Ascorbic acid and α -tocopherol had opposing effects on iron uptake and on MDA production. Thus, the addition of ascorbate resulted in a significant reduction in iron uptake. This effect became more pronounced with time, so that by 24 hours iron uptake with ascorbate was only 27% of the control culture of iron-loaded cells ($P < 0.001$). In contrast, the addition of α -tocopherol resulted in a small but significant increment in iron uptake ranging from 67% at 15 minutes ($P < 0.001$) to 19% at 24 hours ($P < 0.025$).

The effect of ascorbate and α -tocopherol on cellular MDA concentrations was the opposite of their effect on iron uptake. Thus, despite the reduced iron uptake observed in the presence of ascorbate, there was a more than twofold increase in cellular MDA ($P < 0.001$). Likewise, in spite of the moderate increase in iron uptake observed with α -tocopherol, cellular MDA concentrations were markedly reduced between 1 and 24 hours of incubation ($P < 0.001$). To distinguish between an additive as against a synergistic effect on MDA concentrations of iron loading on one hand and hypoxia, α -tocopherol, or ascorbate on the other, the latter variables were tested over periods of 0 to 3 hours without simultaneous iron loading. There was no significant change in the MDA content of untreated control cultures within the study period. In contrast to MDA in control preparations (0.276 $\pm$ 0.032 nmol/mg), the MDA content of cultured cells after 3 hours of hypoxia was 0.513 $\pm$ 0.026 nmol/mg, and after the same period of treatment with ascorbic acid or α -tocopherol 2.949 $\pm$ 0.147 and 0.250 $\pm$ 0.045 nmol/mg respectively. Comparison of these measurements with MDA concentra-

Table III. Lipid peroxidation ratio*

Time (hr)	Iron loading	Iron loading plus hypoxia	Iron loading plus ascorbate	Iron loading plus α -tocopherol
0.25	1	0.48	3.03	0.60
1	1	0.38	2.81	0.37
3	1	0.82	5.74	0.45
24	1	—	7.29	0.13

*Ratio of increase in cellular MDA to percent radioactive iron uptake in experimental group divided by corresponding ratio in iron-loaded cells without additional treatment.

tions after 3 hours of treatment with the same variables in combination with iron loading indicates that their combined effect was clearly not additive but synergistic enhancement or inhibition.

To characterize further the ability of hypoxia, ascorbate, or α -tocopherol to modify the effect of short-term iron loading on MDA formation, the ratio of cellular MDA to percent radioactive iron uptake in treated cultures (iron loading plus ascorbate, α -tocopherol, or hypoxia) was divided by the corresponding ratio in iron-loaded cells without additional treatment (Table III). By definition, this "lipid peroxidation ratio" was 1 in iron-loaded cells without additional treatment. Hypoxia had a modest inhibitory effect on the lipid peroxidation ratio. Ascorbate and α -tocopherol had opposing effects increasing with time. Thus, 24 hours of combined iron loading and ascorbate treatment resulted in a sevenfold increase in MDA as compared with iron loading only, whereas α -tocopherol resulted in an eightfold reduction in the lipid peroxidation ratio, a more than 50-fold difference between the two extremes.

Finally, the effect of ascorbic acid and α -tocopherol on the ability of deferoxamine to remove iron from iron-loaded cells and to alter MDA concentrations in such cells was studied (Table IV). Cultured cells were first exposed to radioactive iron-labeled ferric ammonium citrate at the concentrations used above for 24 hours, washed with normal tissue culture medium, and then treated as specified in Table IV for 3 hours. There was no evidence of spontaneous iron release throughout the duration of the study. Deferoxamine treatment (0.3 mmol) resulted in a reduction of cellular radioactive iron content to 47% of control values within 3 hours. There was no measurable effect of ascorbic acid or α -tocopherol treatment on either spontaneous or deferoxamine-induced iron mobilization. In contrast, both ascorbic acid and α -tocopherol had a profound effect on cellular MDA concentrations. Ascorbate treatment of iron-loaded cells resulted in increased cellular MDA, whereas α -tocopherol treatment resulted in reduced MDA concentrations. Deferoxamine treatment for 3 hours resulted in a 40% reduction in MDA concentrations compared with iron-loaded controls. This

beneficial effect of deferoxamine on MDA production was enhanced by simultaneous α -tocopherol administration, and completely prevented by ascorbate treatment.

DISCUSSION

The pathologic accumulation of iron in chronic transfusional iron overload is associated with increased lipid peroxidation. This has been documented by increased splenic MDA concentrations in patients with thalassemia,¹ as well as in iron-loaded experimental animals in various other tissues such as the liver and pancreas.²⁻⁶ Apart from MDA measurements, methods for documenting increased lipid peroxidation included the demonstration of increased conjugated dienes in tissues or increased excretion of expiratory pentane.^{4,7} The phenomenon of increased lipid peroxidation is not restricted to chronic iron overload inasmuch as it was also demonstrated in short-term iron loading in vivo in rats⁶ or in vitro in hepatic microsomes⁵ or in liposomes.⁸

Although at present no experimental model is available for reproducing the cardiomyopathy associated with chronic iron overload in humans, there are several available systems in which impaired myocardial cell function in short-term iron-loading may be studied. Artman et al.¹⁶ have shown that acute in vivo iron toxicity in rabbits results in reduced stroke volume, cardiac output, right ventricular force, and rate of rise in left ventricular pressure. These authors have also shown that in vitro iron loading of isolated isometrically contracting rabbit atrial strips or papillary muscles results in impaired contractility.¹⁷ Likewise, our group has shown that short-term iron loading of cultured rat myocardial cells results in impaired contractility and altered rhythmicity.¹⁸ In a recent study¹³ we have also shown that short-term iron loading of cultured rat myocardial cells is associated with a rapid increase in lipid peroxidation as indicated by increased cellular MDA concentrations reaching maximal levels after a few hours of iron loading. Although the mechanism whereby increased lipid peroxidation in cultured myocytes results in impaired contractility is at present unknown, in other cell types, such as hepatocytes, iron overload

Table IV. Effect of ascorbic acid and α -tocopherol on deferoxamine-induced iron mobilization and cellular MDA concentrations in iron-loaded cells

Time (hr)	Control	Ascorbate	α -Tocopherol	Deferoxamine	Deferoxamine plus ascorbate	Deferoxamine plus α -tocopherol
Residual ^{59}Fe (%)						
0.5	37.8 $\pm$ 2.0	41.9 $\pm$ 3.8	36.7 $\pm$ 2.1	33.0 $\pm$ 2.6	33.9 $\pm$ 1.7	30.9 $\pm$ 0.9
1.0	39.8 $\pm$ 2.8	39.5 $\pm$ 1.3	35.4 $\pm$ 0.8	29.1 $\pm$ 2.0	27.9 $\pm$ 0.8	27.7 $\pm$ 1.6
3.0	39.2 $\pm$ 2.1	39.2 $\pm$ 0.5	36.1 $\pm$ 1.6	18.6 $\pm$ 0.9	18.3 $\pm$ 0.6	17.2 $\pm$ 1.0
MDA (nmol/mg)						
0.5	3.262 $\pm$ 0.621	4.589 $\pm$ 0.385	2.788 $\pm$ 0.032	2.737 $\pm$ 0.244	1.436 $\pm$ 0.160	1.897 $\pm$ 0.135
1.0	3.404 $\pm$ 0.128	4.269 $\pm$ 0.179	1.917 $\pm$ 0.096	2.545 $\pm$ 0.269	3.442 $\pm$ 0.410	1.731 $\pm$ 0.096
3.0	2.654 $\pm$ 0.352	3.994 $\pm$ 0.109	1.301 $\pm$ 0.154	1.609 $\pm$ 0.237	4.865 $\pm$ 0.212	0.763 $\pm$ 0.218

Each value represents mean $\pm$ SD of four studies.

was shown to interfere with adenosine diphosphate-stimulated mitochondrial oxidative metabolism.¹⁹ Other possible targets of iron toxicity are the lysosomal membrane^{3,20} or, in the case of myocardial cells, the sarcolemmal membrane. It should be emphasized that the conditions used in our cultured myocardial cell model are comparable with acute, and not chronic, iron poisoning in the concentrations of non-transferrin-bound iron used and the short duration of exposure. Similarly, the high concentrations of ascorbic acid and α -tocopherol used were designed to simulate conditions in previous *in vitro* studies^{5,8} and do not represent physiologic *in vivo* conditions.

Because of the central role of lipid peroxidation in the pathogenesis of iron-induced tissue damage, it is important to define conditions that may modify the peroxidative capacity of iron. Several previous observations gave reason to believe that ascorbate may have a profound effect on the potential toxicity of iron. Nienhuis²¹ commented on the rapid deterioration of hemochromatotic cardiomyopathy in some patients with thalassemia given ascorbic acid to enhance the efficacy of deferoxamine mesylate treatment.²² Wills⁵ has shown in a microsomal system that the rate of lipid peroxidation is strongly stimulated by ascorbic acid in the presence of ferrous iron. Recently, O'Connell et al.⁸ have shown in liposomes that whereas ferritin or hemosiderin per se had no effect on lipid peroxidation, the production of MDA was strikingly increased by the addition of ascorbic acid to the system. The postulated mechanism of this effect was mobilization of insoluble ferric iron stored in ferritin by the formation of a highly soluble ferric ascorbate complex¹² and stimulation of hydroxyl radical formation via the Fenton reaction.⁹ Both of our present findings, namely, the reduced accumulation of ferric ammonium citrate in myocardial cells and the marked increase in MDA concentrations, are compatible with the above mechanisms. Reduction of ferric to ferrous iron and the formation of the highly

soluble ferrous-ascorbate complex may have resulted in reduced radioactive iron uptake as well as mobilization of ferritin or hemosiderin iron already within cells. Likewise, the increased formation of ferrous iron produced by the interaction of ascorbate and ferric iron complexes resulted most probably in the accelerated conversion of H_2O_2 into $\cdot\text{OH}$ and OH^- by the Fenton reaction, ultimately resulting in increased lipid peroxidation.

The *in vivo* antioxidant effect of α -tocopherol has been elegantly demonstrated by Dillard et al.⁷ in iron-loaded rats by showing a 92% depression of pentane excretion in exhaled air after the addition of α -tocopherol to their diet. In liposomes, O'Connell et al.⁸ have shown that incorporation of α -tocopherol into the liposomal membrane inhibited completely iron-induced MDA production by its chain-breaking antioxidant action. Our present studies in myocardial cells have shown that, similar to other experimental models, iron-induced lipid peroxidation, as manifested by cellular MDA concentrations, is drastically reduced by the presence of α -tocopherol. It would be appropriate at this point to caution against equating our measurements of tissue MDA concentrations with rates of lipid peroxidation. Concentrations of MDA are the result of a dynamic equilibrium between rates of production and breakdown. *In vivo* and *in vitro* studies have shown that MDA is rapidly processed by mitochondrial aldehyde dehydrogenase followed by decarboxylation and the production of CO_2 and acetate.²³ Hence, any change in tissue MDA concentrations may be the result of altered MDA production as well as catabolism. Nevertheless, our conclusions that ascorbate enhances and α -tocopherol inhibits myocardial lipid peroxidation are strongly supported by previous studies^{7,8} in which these effects were confirmed independently by other manifestations of lipid peroxidation such as the quantitation of expiratory pentane or direct measurements of membrane lipid composition. A further limitation of using

total tissue MDA measurements is the inability to distinguish internal peroxidation of cellular organelles from the effects of extracellular iron on the cellular (sarcolemmal) membrane.

In view of *in vivo* evidence for the ability of ascorbic acid to enhance the iron-chelating ability of deferoxamine,^{24,25} and a possible inhibitory effect of α -tocopherol on *in vivo* iron chelation,²⁶ we wished to examine the ability of these two compounds to modify the effect of deferoxamine *in vitro*. Neither one of these compounds had any effect on the ability of deferoxamine to remove iron from myocardial cells. This apparent contradiction between *in vivo* and *in vitro* observations may be possibly explained by an indirect effect of vitamin C and E on iron mobilization elsewhere such as the release of iron from the reticuloendothelial system. In contrast, both ascorbic acid and α -tocopherol had a pronounced effect on the ability of deferoxamine to alter cellular MDA concentrations. Combined ascorbic acid and deferoxamine treatment resulted in a sharp increase in MDA concentrations in contrast to the reduction in MDA observed with deferoxamine treatment alone. Conversely, simultaneous α -tocopherol and deferoxamine treatment resulted in a further reduction in cellular MDA as compared with that observed with deferoxamine only. These observations suggest that ascorbic acid or α -tocopherol treatment may exert a profound modifying effect on the ability of deferoxamine to prevent or reverse iron-induced myocardial membrane lipid peroxidation.

Our interest in the effect of hypoxia on iron uptake and lipid peroxidation arose from the unique observation of Necheles et al.²⁷ in congenitally anemic mice that coexistent iron loading and hypoxia resulted in increased rates of myocardial iron accumulation. Our *in vitro* studies are entirely in line with these observations. Coexistent iron loading and hypoxia resulted in a twofold to threefold increase in myocardial iron uptake in cultured cells. This increased uptake, however, did not cause a corresponding enhancement of MDA production. We assume that the failure to increase MDA production is probably the result of a temporary reduction in free radical formation under hypoxic conditions.

It is important to be aware of the fundamental difference between *in vitro* and *in vivo* conditions and of the significant difference in the clinical manifestations of acute as against chronic toxicity. Nevertheless, the present observations on the modifying effects of hypoxia, ascorbate, or α -tocopherol treatment on myocardial iron uptake, toxicity, and chelation may be of some relevance to iron overload in transfusional siderosis.

The accelerated rate of myocardial iron deposition in

hypoxia may be regarded as a further argument in favor of maintaining near-normal hemoglobin levels in patients with thalassemia major and other iron-loading anemias requiring continued transfusion therapy. The slight increase in transfusional requirements needed to maintain adequate hemoglobin levels is more than balanced by the beneficial effects of correcting anemia and avoiding further organ damage.

The present findings indicating a marked increase in myocardial lipid peroxidation after ascorbate therapy underline the controversy regarding vitamin C supplementation in thalassemia.^{21,22} Ascorbate deficiency is a common complication of iron overload caused by the accelerated conversion of ascorbate to oxalic acid in the presence of excess tissue iron.²⁴ Furthermore, ascorbate supplementation potentiates the *in vivo* effect of deferoxamine²⁵ probably by enhancing storage iron mobilization and by increasing the chelatable iron pool. However, in view of the aggravation of hemochromatotic cardiomyopathy observed in some patients receiving ascorbic acid supplementation to enhance responsiveness to deferoxamine therapy,²² such supplementation should be carried out with extreme caution if at all.

The therapeutic use of α -tocopherol for protecting membranes against lipid peroxidation has been advocated in the past, with some success reported in the management of hereditary hemolytic anemia.²⁸ To our knowledge, the present report is the first one in which the protective effect of α -tocopherol against lipid peroxidation in myocardial cells has been directly demonstrated. Our previous study of iron-loaded myocardial cells in culture as well as our present results indicate that deferoxamine is capable of reducing myocardial MDA concentrations in direct proportion to the amounts of iron removed.¹³ Thus, at least two methods of protecting myocardial cells from increased lipid peroxidation are available: removal of excess iron by a metal chelator, or preventing the propagation of lipid peroxidation by the chain-breaking antioxidant α -tocopherol. In contrast to deferoxamine therapy, which is now widely used in the management of transfusional iron overload, the therapeutic potential of α -tocopherol in hemochromatosis has not yet been explored.

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